

Home Assignment_Test case Generator_

1. Update the sample prompt given as needed to create a test case generator application
2. Create a Test case Generator folder in desktop or in any path

E:\Vidhya_Assignment link\August 29th\Home Assignment\Testcase_Generator

2. Give the updated prompt in Chat GPT to get the architecture file to build the application

Architecture file given by Chat GPT :

Absolutely. Since we've already completed all 20 architecture questions, below is the complete `Architecture.md` for your ICE POT – AI Test Case Generator. This is designed as an enterprise-grade solution using your chosen stack (React, Node.js + TypeScript, Express, MongoDB Vector DB, Mistral Embeddings, OpenAI/Groq/Anthropic) and incorporates all the decisions you made.

> File Name: `Architecture.md`

ICE POT – AI Test Case Generator

Enterprise Solution Architecture

* Version: 1.0

* Architecture Style: Serverless + Multi-Agent

* Frontend: React + TypeScript

* Backend: Node.js + TypeScript + Express

* Vector Database: MongoDB

* Embedding Model: Mistral AI

* LLM Providers: OpenAI, Groq, Anthropic (User Selectable)

Table of Contents

1. Executive Summary

2. Architecture Decisions

3. High-Level Architecture

4. End-to-End Workflow

5. Frontend Architecture

6. Backend Architecture

7. Multi-Agent Architecture

8. Ingestion Pipeline

9. Retrieval Pipeline

10. MongoDB Data Model

11. REST API Endpoints

12. SSE Events

13. Traceability

14. Release Impact Engine

15. Confidence Engine

16. Security

17. Observability

18. Project Folder Structure

19. Implementation Roadmap

1. Executive Summary

ICE POT is an AI-powered Test Case Generator that ingests multiple software engineering artifacts, converts them into semantic embeddings, retrieves relevant context using hybrid search, and generates high-quality test cases with Human-in-the-Loop approval.

Supported Inputs

|

Input

|

Supported

|
| --- | --- |
|

BRD

|

|
|

User Stories

|

|
|

Jira

|



|

|

Azure DevOps

|



|

|

TestRail

|



|

|

Xray

|



|

|

Zephyr

|



|

|

Figma

|



|

|

Confluence/Wiki

|



|

|

Release Notes

|



|

|

Excel

|



|

|

PDF

|



|

|

Grooming Recordings

|



|

|

Git Repository

|



|

|

Swagger/OpenAPI

|



|

|

Defect Database

|



|

2. Architecture Decisions

|

Question

|

Decision

|

| --- | --- |

|

Architecture

|

Serverless

|

|

Authentication

|

Google SSO

|

|

Ingestion

|

Hybrid

|

|

Processing

|

User-triggered

|

|

Chunking

|

Hybrid

|

|

Isolation

|

Workspace-based

|

|

LLM

|

User selects

|

|

API

|

REST + SSE

|

|

Retrieval

|

BM25 + Vector + Reranking

|

|

Prompt Management

|

Hybrid

|

|

Review

|

Draft → Review → Approved

|

|

Export

|

Manual

|

|

Swagger

|

Upload + URL

|

|

Recording

|

Meeting Intelligence

|

|

Code

|

Knowledge Graph

|

|

Confidence

|

Explainable

|

|

Traceability

|

Complete

|

|

Release

|

Intelligent Impact

|

|

AI Design

|

Multi-Agent

|

|

Monitoring

|

Enterprise Governance

|

3. High-Level Architecture

4. End-to-End Workflow

...

User Login



Select Workspace



Select Project



Upload Sources / Connectors



Click "Create Embeddings"



Content Extraction



Hybrid Chunking



Mistral Embeddings



MongoDB



User Query



Hybrid Retrieval



Selected LLM



Draft Test Cases



Review



Approve / Reject / Modify



Export

...

5. Frontend Architecture

Technology

- * React

- * TypeScript

- * React Router

- * Redux Toolkit / React Query

- * Server-Sent Events

Major Screens

* Login

* Dashboard

* Workspace

* Projects

* Source Upload

* Connectors

* Ingestion Status

* Ask ICE POT

* Test Case Generator

* Review Screen

* Traceability Matrix

* Release Dashboard

* Confidence Details

* Prompt Management

- * Audit Dashboard

6. Backend Architecture

Services

- * Auth Service

- * Workspace Service

- * Project Service

- * Source Service

- * Ingestion Service

- * Embedding Service

- * Retrieval Service

- * Generation Service

- * Review Service

- * Traceability Service

* Release Service

* Confidence Service

* Export Service

* Audit Service

All services are stateless and suitable for serverless deployment.

7. Multi-Agent Architecture

|

Agent

|

Responsibility

|

| --- | --- |

|

Ingestion

|

Parse documents

|

|

Embedding

|

Create Mistral embeddings

|

|

Retrieval

|

Hybrid search

|

|

Code Intelligence

|

Build knowledge graph

|

|

Test Generation

|

Generate test cases

|

|

Traceability

|

Maintain relationships

|

|

Release Impact

|

Analyze releases

|

|

Confidence

|

Calculate explainable score

|

|

Review

|

Approval workflow

|

8. Ingestion Pipeline

Manual Upload

...

Upload



Validation



Content Extraction



Normalization



Chunking



Embeddings



MongoDB

...

Connector Flow

...

Jira / ADO / Git / Swagger



Connector



Common Artifact



Embeddings

...

9. Retrieval Pipeline

This follows your exact requirement.

```
%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22204%22%20x2%3D%22410%22%20y2%3D%22224%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22272%22%20x2%3D%22210%22%20y2%3D%22310%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22272%22%20x2%3D%22610%22%20y2%3D%22310%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22210%22%20y1%3D%22366%22%20x2%3D%22410%22%20y2%3D%22392%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22610%22%20y1%3D%22366%22%20x2%3D%22410%22%20y2%3D%22392%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22440%22%20x2%3D%22410%22%20y2%3D%22460%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22508%22%20x2%3D%22410%22%20y2%3D%22528%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22576%22%20x2%3D%22410%22%20y2%3D%22596%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22644%22%20x2%3D%22410%22%20y2%3D%22664%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22712%22%20x2%3D%22410%22%20y2%3D%22732%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3Cline%20x1%3D%22410%22%20y1%3D%22780%22%20x2%3D%22410%22%20y2%3D%22800%22%20marker-  
end%3D%22url\(%23arrow\)%)%22%2F%3E%3C%2Fg%3E%3Cg%20font-size%3D%2216%22%20text-  
anchor%3D%22middle%22%3E%3Ctext%20x%3D%22410%22%20y%3D%2249%22%3EUser%20Query%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22117%22%3EQuery%20Normalization%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22185%22%3EAbbreviation%20Expansion%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22253%22%3ESynonym%20Expansion%3C%2Ftext%3E%3Ctext%20x%3D%22210%22%20y%3D%22343%22%3EBM25%20Search%3C%2Ftext%3E%3Ctext%20x%3D%22610%22%20y%3D%22343%22%3EVector%20Search%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22421%22%3EMerge%20Results%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22489%22%3EReranking%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22557%22%3EDeduplication%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22625%22%3ESummarization%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22693%22%3EPrompt%20Construction%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22761%22%3ESelected%20LLM%3C%2Ftext%3E%3Ctext%20x%3D%22410%22%20y%3D%22829%22%3EGenerated%20Test%20Cases%3C%2Ftext%3E%3C%2Fg%3E%3C%2Fsvg%3E)

# 10. MongoDB Collections

|  
Collection

|

Purpose

|  
| --- | --- |  
|

workspaces

|

Workspace info

|  
|

projects

|

Project info

|

|

users

|

Users

|

|

sources

|

Uploaded sources

|

|

artifacts

|

Normalized artifacts

|

|

artifactChunks

|

Chunks

|

|

embeddings

|

Vector storage

|

|

apis

|

Swagger APIs

|

|

codeEntities

|

Code graph

|

|

testCases

|

Test cases

|

|

testCaseVersions

|

Versions

|

|

comments

|

Review comments

|

|

releases

|

Releases

|

|

traceability

|

Relationships

|

|

confidence

|

Confidence

|

|

auditLogs

|

Audit

|

|

prompts

|

Prompt templates

|

## # 11. REST API Endpoints

### ## Authentication

|

Method

|

Endpoint

|

| --- | --- |

|

GET

|

`/api/v1/auth/google`

|

|

GET

|

`/api/v1/auth/callback`

|

|

POST

|

`/api/v1/auth/logout`

|

|

GET

|

`/api/v1/auth/me`

|

## Workspace APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/api/v1/workspaces`

|

|

GET

|

`/api/v1/workspaces`

|

|

GET

|

`/api/v1/workspaces/{id}`

|

|

PATCH

|

`/api/v1/workspaces/{id}`

|

|

DELETE

|

`/api/v1/workspaces/{id}`

|

## ## Project APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/api/v1/workspaces/{workspaceId}/projects`

|

|

GET

|

`/api/v1/workspaces/{workspaceId}/projects`

|

|

GET

|

`/api/v1/projects/{projectId}`

|

|

PATCH

|

`/api/v1/projects/{projectId}`

|

|

DELETE

|

`/api/v1/projects/{projectId}`

|

## Source Management

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/api/v1/projects/{projectId}/sources/upload`

|

|

GET

|

`/api/v1/projects/{projectId}/sources`

|

|

DELETE

|

`/api/v1/projects/{projectId}/sources/{sourceId}`

|

## Connector APIs

|

Source

|

Endpoint

|

| --- | --- |

|

Jira

|

`/connectors/jira`

|

|

ADO

|

`/connectors/ado`

|

|

Confluence

|

`/connectors/confluence`

|

|

Git

|

`/connectors/github`

|

|

TestRail

|

`/connectors/testrail`

|

|

Xray

|

`/connectors/xray`

|

|

Zephyr

|

`/connectors/zephyr`

|

|

Figma

|

`/connectors/figma`

|

|

Defects

|

`/connectors/defects`

|

## Swagger APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/swagger/upload`

|

|

POST

|

`/swagger/url`

|

|

GET

|

`/swagger/apis`

|

## Embedding APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/embeddings`

|

|

GET

|

`/embeddings/status`

|

|

GET

|

`/embeddings/{jobId}`

|

## Retrieval APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/search`

|

|

POST

|

`/retrieve`

|

## ## Test Generation APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/test-cases/generate`

|

|

GET

|

`/test-cases`

|

|

GET

|

`/test-cases/{id}`

|

|

PATCH

|

`/test-cases/{id}`

|

## Review APIs

|

Action

|

Endpoint

|  
| --- | --- |  
|

Approve

|

`/approve`

|  
|

Reject

|

`/reject`

|  
|

Modify

|

`/modify`

|

|

Versions

|

`/versions`

|

|

Comments

|

`/comments`

|

## Traceability APIs

|

Method

|

Endpoint

|

| --- | --- |

|

GET

|

`/traceability`

|

|

GET

|

`/coverage`

|

|

GET

|

`/orphans`

|

|

POST

|

`/impact-analysis`

|

## Release APIs

|

Method

|

Endpoint

|  
| --- | --- |  
|

POST

|

`/releases`

|  
|

POST

|

`/releases/{id}/analyze`

|  
|

GET

|

`/releases/{id}/impact`

|

## Confidence APIs

|

Method

|

Endpoint

|

| --- | --- |

|

GET

|

`/confidence`

|

|

GET

|

`/evidence`

|

## Export APIs

|

Method

|

Endpoint

|

| --- | --- |

|

POST

|

`/exports`

|

|

GET

|

`/exports/{id}`

|

## # 12. Server-Sent Events

ICE POT uses SSE for live updates.

Events:

\* `INGESTION\_STARTED`

\* `INGESTION\_PROGRESS`

\* `EMBEDDING\_PROGRESS`

\* `EMBEDDING\_COMPLETED`

\* `RETRIEVAL\_PROGRESS`

\* `GENERATION\_PROGRESS`

\* `GENERATION\_COMPLETED`

\* `REVIEW\_UPDATED`

### # 13. Complete Traceability

%%22%2F%3E%3Cline%20x1%3D%22260%22%20y1%3D%22356%22%20x2  
%3D%22260%22%20y2%3D%22380%22%20marker-  
end%3D%22url\(%23arrow2\)%%22%2F%3E%3Cline%20x1%3D%22260%22%20y1%3D%22428%22%20x2  
%3D%22260%22%20y2%3D%22452%22%20marker-  
end%3D%22url\(%23arrow2\)%%22%2F%3E%3Cline%20x1%3D%22260%22%20y1%3D%22500%22%20x2  
%3D%22260%22%20y2%3D%22524%22%20marker-  
end%3D%22url\(%23arrow2\)%%22%2F%3E%3C%2Fg%3E%3Cg%20font-size%3D%2216%22%20text-  
anchor%3D%22middle%22%3E%3Ctext%20x%3D%22260%22%20y%3D%2249%22%3E%3C%2Ftext  
%3E%3Ctext%20x%3D%22260%22%20y%3D%22121%22%3E%3CUser%20Story%3C%2Ftext%3E%3Ctext%20  
x%3D%22260%22%20y%3D%22193%22%3E%3CAcceptance%20Criteria%3C%2Ftext%3E%3Ctext%20x%3D%  
22260%22%20y%3D%22265%22%3E%3ESwagger%20API%3C%2Ftext%3E%3Ctext%20x%3D%22260%22%20  
y%3D%22337%22%3E%3CSource%20Code%3C%2Ftext%3E%3Ctext%20x%3D%22260%22%20y%3D%22409  
%22%3E%3CGenerated%20Test%20Cases%3C%2Ftext%3E%3Ctext%20x%3D%22260%22%20y%3D%22481%  
22%3E%3CApproved%20Test%20Cases%3C%2Ftext%3E%3Ctext%20x%3D%22260%22%20y%3D%22553%22  
%3E%3CDefects%3C%2Ftext%3E%3C%2Fg%3E%3C%2Fsvg%3E)

Capabilities:

- \* Requirement coverage

- \* API coverage

- \* Code linkage

- \* Orphan detection

- \* Coverage gaps

- \* Impact analysis

# 14. Release Impact Engine

Pipeline:

...

Release Notes



Change Detection



Requirement Mapping



Swagger Diff



Code Impact



Existing Test Cases



Regression Recommendation

...

Outputs:

- \* New test cases

- \* Modified test cases

- \* Deprecated tests

- \* Regression suite

\* Coverage gaps

## # 15. Explainable Confidence Engine

Example:

|

Factor

|

Score

|

| --- | --- |

|

Retrieval

|

96%

|

|

Coverage

|

95%

|

|

Reranking

|

93%

|

|

Consistency

|

90%

|

|

Prompt

|

94%

|

|

Generation

|

88%

|

Overall Confidence: 92%

Evidence:

\* BRD Section 4.2

\* User Story

\* Swagger Endpoint

- \* Code Reference

- \* Grooming Transcript

## # 16. Security

- \* Google OAuth

- \* RBAC

- \* Workspace isolation

- \* Project isolation

- \* JWT/session validation

- \* Encrypted secrets

- \* API rate limiting

- \* File validation

- \* Prompt injection protection

- \* Audit logs

## # 17. Enterprise Observability

Monitor:

- \* API latency

- \* Token usage

- \* LLM provider

- \* Prompt version

- \* Cost

- \* Errors

- \* User actions

- \* Approvals

- \* Exports

Every request receives:

...

X-Correlation-ID

...

## # 18. Project Folder Structure

...

ice-pot/

1

└─ apps/

```
| └─ web/
```

```
| | |└─ src/
```

| | | | — components/

| | | | — pages/

```
| | | |— hooks/
```

```
| | | |—services/
```

| | | | — store/

| | | | — types/

```
| | | └─ utils/
```

```
| | └─ package.json
```

11

| └─ api/

[illegible]

```
| | |└─ routes/
```

| | | — controllers/

| | └─ services/

| | | — agents/

| | | — adapters/

| | | — providers/

```
| | ├── middleware/
```

| | ├── models/



```
|
|├─ tests/
| |├─ unit/
| |├─ integration/
| |├─ contract/
| |├─ e2e/
| └─ ai-evaluation/
|
|├─ docs/
| |├─ architecture/
| |├─ api/
| |├─ agents/
| └─ deployment/
|
|├─ package.json
|├─ tsconfig.json
|├─ README.md
└─ Architecture.md
...

```

## # 19. Implementation Roadmap

### ## Phase 1 (MVP)

\* Google SSO

\* Workspace & Projects

- \* File Upload

- \* BRD/User Story ingestion

- \* Mistral Embeddings

- \* MongoDB Vector Search

- \* Hybrid Retrieval

- \* Test Case Generation

- \* Draft → Review → Approved

- \* Excel/PDF/CSV/Markdown Export

## ## Phase 2

- \* Jira

- \* ADO

- \* Confluence

- \* Swagger

- \* API Test Generation

- \* Traceability

- \* Confidence Engine

## ## Phase 3

- \* Git Knowledge Graph

- \* TestRail

- \* Xray

- \* Zephyr

- \* Defect Integration

- \* Release Intelligence

## ## Phase 4

- \* Voice Chat

- \* AI Analytics

- \* Advanced Governance

- \* Automated Regression Recommendation

## # Final Architecture Summary

The final ICE POT architecture combines:

- \* Serverless Architecture

- \* React + Node.js + TypeScript + Express

- \* MongoDB Vector Database

- \* Mistral AI Embeddings

- \* User-selectable OpenAI, Groq, and Anthropic

- \* Hybrid BM25 + Vector Retrieval

- \* Multi-Agent AI Architecture

- \* Human-in-the-Loop Review

- \* Explainable Confidence Scoring

- \* Repository Knowledge Graph

\* Complete Traceability

\* Release Impact Intelligence

\* Enterprise Observability

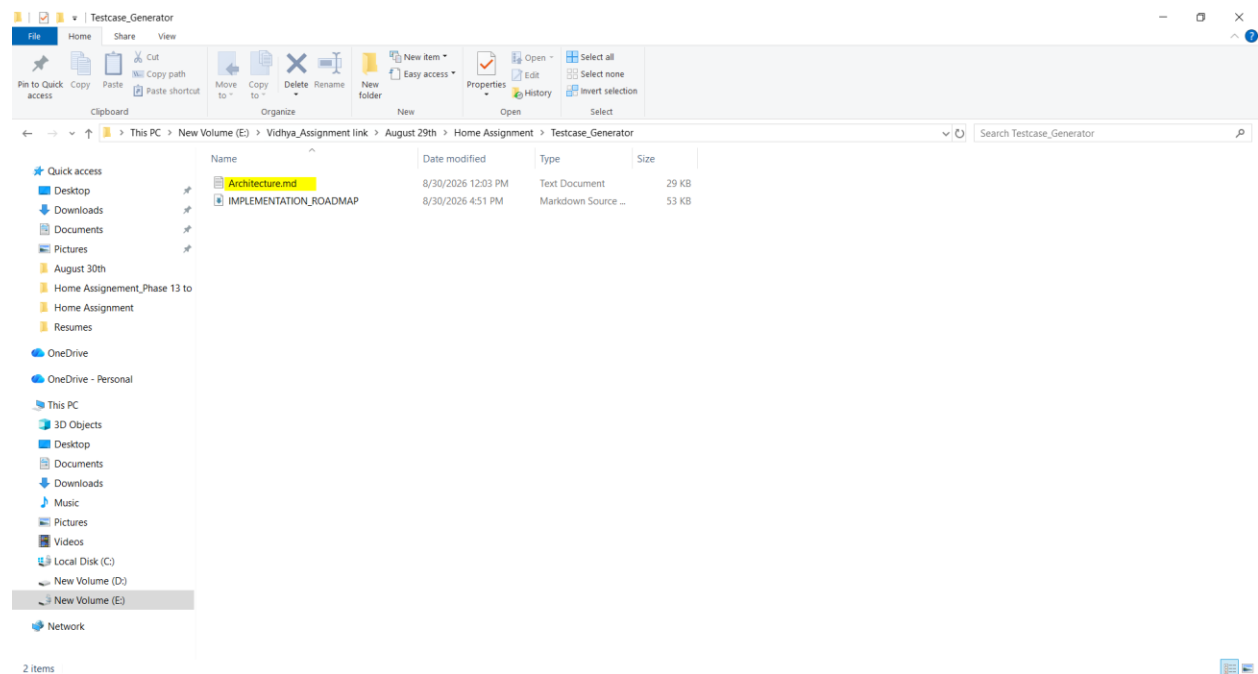
This architecture is modular, scalable, and designed so future integrations (Jira, ADO, TestRail, Xray, Zephyr, additional LLMs, and more exporters) can be added through adapters without changing the core platform.

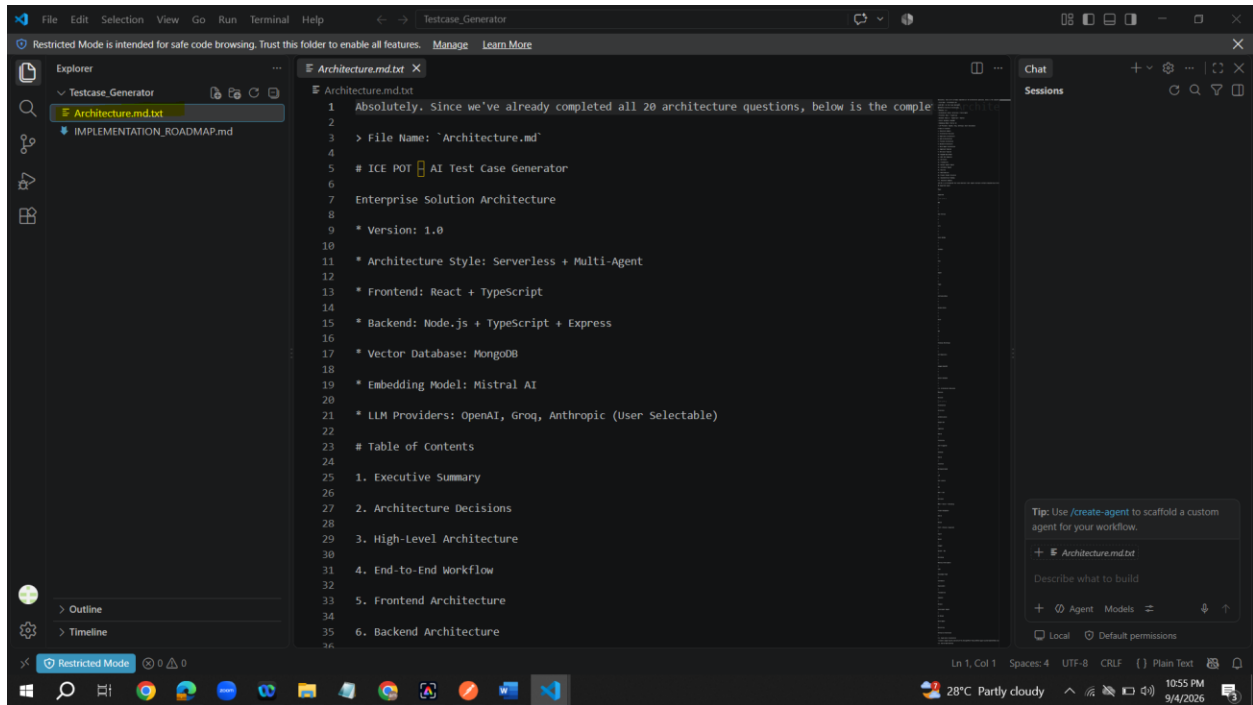
3. Copy the architecture file given by Chat GPT in a notepad and save it inside Test case Generator folder in desktop or in the path you have the 'Testcase Generator' folder

4. Open VS code.File-New Window.File-Open folder

**C:\Users\rta\Desktop\Testcase\_Generator or**

**E:\Vidhya\_Assignment link\August 29th\Home Assignment\Testcase\_Generator**





5. In copilot within VS code paste the architecture file contents given by Chat GPT and give the below prompt. Phase by phase implementation plan will be created

**Analyse and segregate this architecture into phase by phase implementation and generate the .md file for phase wise implementation**

